

INFLUENCE OF INTESTINAL BIORELEVANT FLUIDS ON A NOVEL CO-CULTURE CELL MODEL ABLE TO MIMIC VARIABLE PERMEABILITIES OF THE HUMAN INTESTINE

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1 - PURPOSE

The study focuses on the development of a new model of intestinal barrier using a coculture of Caco-2 absorptive cells and mucin-secreting HT29-MTX goblet cells¹. In a previous work, paracellular transport and P-gp mediated efflux were adjusted according to the human intestine parameters. To mimic accurately *in-vivo* conditions, the choice of the incubation medium used for the permeability studies was investigated. Compatibility of biorelevant fluids so-called Fasted State Simulated Intestinal Fluid (FaSSIF) and Fed State Simulated Intestinal Fluid (FeSSIF) with the co-culture was evaluated and compared with *in-vitro* models of intestinal barrier including the caco-2 monolayer.

2 - METHODS

• Cell culture conditions

Cocultures were performed using Transwell®-24 well 0.33 cm² permeable support with a polycarbonate membrane. Caco-2 and HT29-MTX were used from passage 64 to 80 and passage 33 to 42, respectively. Caco-2 were seeded at densities of 5,000 cells per well at day 0 whereas HT29-MTX were added at a density of 10,000 cells per well at day 3. Monolayers were used for transport experiments between day 21 and 30.

• Confocal microscopy

Monolayers cultivated on Transwell® were washed three times with DPBS. Cells were fixed with 1% paraformaldehyde during 10 minutes and washed with DPBS. Monolayers were blocked with 1% BSA and 50% goat serum in the apical compartment for 30 minutes. After three washings, anti-human P-gp mouse primary antibody was incubated with the monolayers at 30 µg/ml for 1hr. Controls were performed using a mouse IgG2a isotype. Alexa Fluor 488-conjugated anti-mouse antibody was added into the apical compartment at 10 µg/ml for 30 minutes.

• Biorelevant fluids preparation (FaSSIF and FeSSIF)

Compositions of FaSSIF and FeSSIF were summarized in table 1. They were used within 24hrs after preparation. Amounts of NaTC and Pancreatin was 2-fold lower than the original composition of FeSSIF-V2².

Table 1: Composition of transport Buffer (TB) and biorelevant media (FeSSIF and FaSSIF). *Italic values correspond to the blank FeSSIF.*

	Transport Buffer	FaSSIF	FeSSIF
Hank's balance Salt Solution	500 mL		
D-(+)-glucose	25 mM		
HEPES	10 mM		
Sodium chloride		68.62 mM	125.5 mM
Calcium chloride		-	5 mM
Maleic acid		19.12 mM	55.05 mM
Sodium hydroxide		34.8 mM	81.65 mM
Sodium oleate		-	0.8 mM
Glycerol mono-oleate		-	5 mM
Sodium taurocholate (NaTC)		3 mM	5 mM
Phosphatidylcholine		0.2 mM	2 mM
Pancreatin			50 U/mL
Sodium hydroxide 1N	q.s. ad pH 57.4	q.s. ad pH 6.5	q.s. ad pH 5.8
Deionized water pH	7.4	q.s. ad 0.1 L 6.5	q.s. ad 0.1 L 5.8

• Incubation studies

FaSSIF and FeSSIF were added in the apical compartment and the transport buffer (TB) in the basolateral compartment. TB in the apical compartment was used as a control. Paracellular permeability was evaluated by measuring the transport of Lucifer Yellow (LY) from apical to basolateral compartments. The determination of the transepithelial electrical resistance (TEER) was carried out using the Millicell Electrical Resistance System before and after the transport studies.

3 - RESULTS AND DISCUSSION

Distinct islet formation for Caco-2 and HT29-MTX within the coculture was showed in Figure 1. Caco-2/HT29-MTX coculture formed multilayers in contrast to monolayers Caco-2. In Figure 2, TEER recovery in the co-culture after a 2hr-incubation of different compositions of FeSSIF showed a synergistic toxic effect by combining 10mM NaTC with 50 or 100 U/ml of pancreatin. Association of 5 mM NaTC with pancreatin at 50 or 100 U/ml were compatible with the co-culture. A toxicity study of FaSSIF and FeSSIF were then performed with different *in-vitro* models of intestinal barrier including the Caco-2 monolayer, considered as the gold standard. In figure 3, a 2hr-incubation of FeSSIF led to a drastic decrease of TEER in the caco-2 monolayer whereas this effect was not observed for the co-culture and HT29-MTX cells. This was consistent with the LY transport study (Figure 4).

FeSSIF enhanced significantly the LY transport across the Caco2 monolayer compared with TB. This high paracellular permeability suggests a disruption of tight junctions after exposure to FeSSIF. TEER recovery after incubations with TB and FaSSIF were in contrast in the same range, independently of the cell line. In addition, FaSSIF medium did not modify the LY permeability compared with the transport buffer, demonstrating a compatibility of FaSSIF with the Caco-2 model (figure 5).

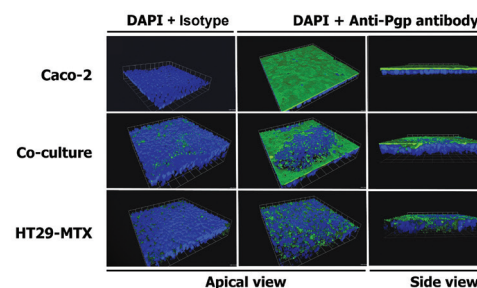


Figure 1: Observation of the distribution of Caco-2 and HT29-MTX in the coculture by confocal microscopy.

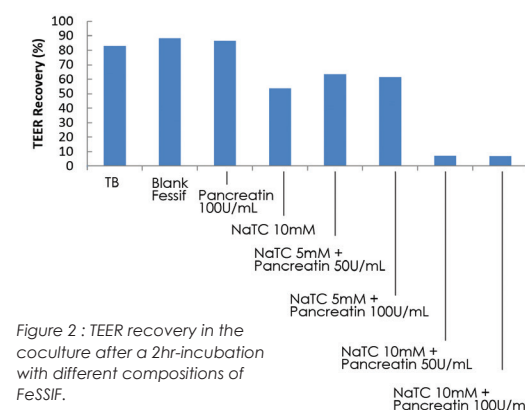


Figure 2 : TEER recovery in the coculture after a 2hr-incubation with different compositions of FeSSIF.

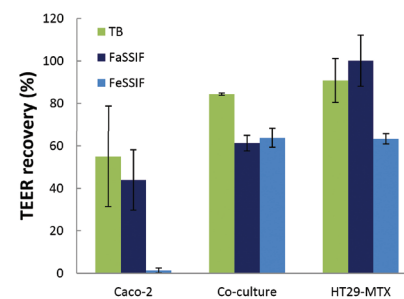


Figure 3 : TEER recovery in the three cell models after a 2hr-incubation with TB, FaSSIF and FeSSIF.

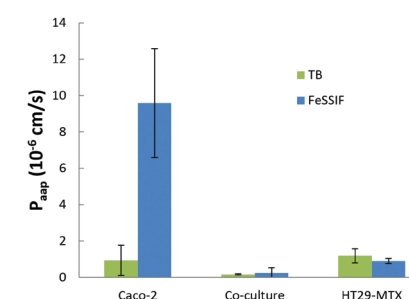


Figure 4: Papp of Lucifer Yellow in TB and FeSSIF after a 2hr-incubation with Caco-2, co-culture and HT29-MTX.

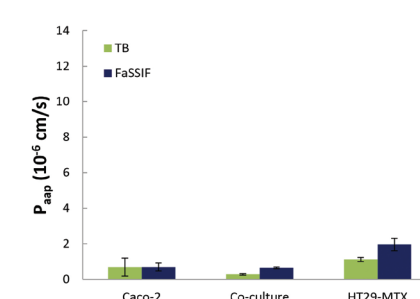


Figure 5: Papp of Lucifer Yellow in TB and FaSSIF after a 2hr-incubation with Caco-2, co-culture and HT29-MTX.

4 - CONCLUSIONS

In addition to the relevancy of the cell model, the choice of biorelevant fluids is a critical point to predict accurately the intestinal permeability of drug candidates. The presence of bile salts, lecithin and enzymes have to be considered due to their role in the solubilization and the metabolism mechanisms of drugs as well as in their interaction with some transporters. In this study, we demonstrated the compatibility of the coculture with biorelevant fluids. In contrast to Caco-2 model, integrity of the co-culture was not affected after a 2hr-exposure to either FaSSIF or FeSSIF.

5 - REFERENCES

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